

Integrating Polyspace Bug Finder and Polyspace Access

<https://www.mathworks.com/help/bugfinder/gs/run-bug-finder-on-server.html>

Prerequisites

Make sure all steps in license_config.pdf are completed

Adding polyspace command to PATH

```
sudo nano ~/.bashrc
```

Add this line to end of the file

```
export PATH="$PATH:/usr/local/Polyspace_Server/R2025b/polyspace/bin
```

Save and exit

Note: If command is not found, either add them to PATH or navigate to the directory and use ./ to execute the command

Start-up Cluster Dashboard

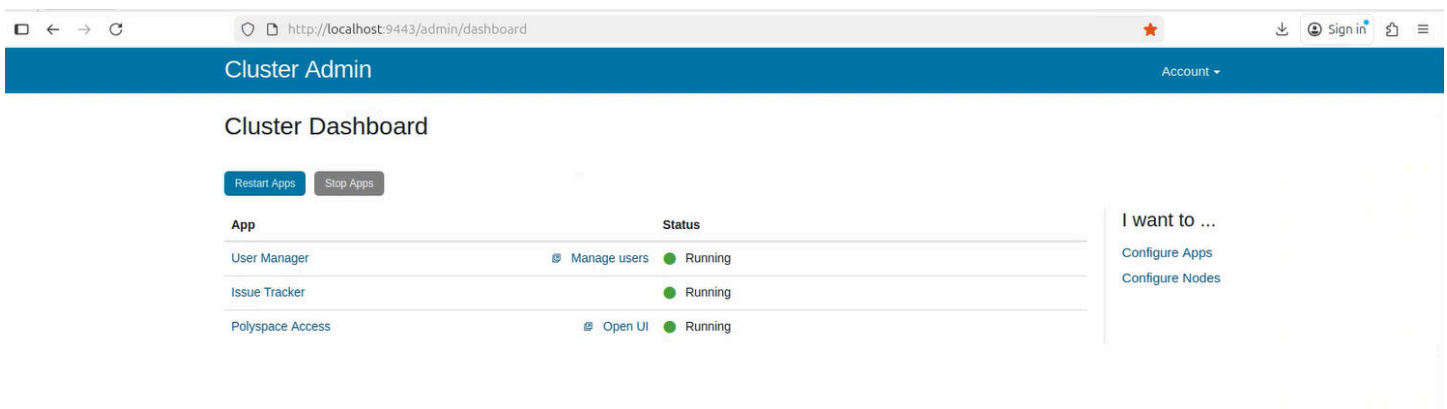
List all running docker container and make sure it's running

```
docker ps
```

Navigate to polyspace access directory in Ubuntu. Launch Cluster Admin interface

```
./admin-docker-agent
```

Open Cluster Dashboard from browser, make sure all 3 status are running



Check Polyspace Bug Finder Server Installation

In Ubuntu, navigate to *polyspaceserverroot*\polyspace\bin. Here, *polyspaceserverroot* is the Polyspace Bug Finder Server installation folder

```
./polyspace-bug-finder-server -help
```

```
polyspace-server-06@polyspace-server-06-VMware-Virtual-Platform:/usr/local/Polyspace_Server/R2025b/polyspace/bin$ ./polyspace-bug-finder-server -help
Syntax: polyspace-bug-finder-server [options]
Bug Finder Analysis
-autosar-cpp14 [all|automated|from-file|required]
-cert-c [all|all-rules|from-file|publish-2016]
-cert-cpp [all|from-file]
-checkers string1[,string2[,...]]
-checkers-activation-file file
-checkers-selection-file file
-code-metrics
-custom-rules [from-file]
-cwe [all|cwe-658-659|all-exact-checkers|from-file]
-disable-checkers string1[,string2[,...]]
-do-not-generate-results-for [all-headers|include-folders|
                             custom=file1,dir1...]
-fast-analysis
-generate-results-for [all-headers|source-headers|custom=file1,dir1...]
-guidelines [all|his|from-file]
-incremental
-iso-17961 [all|decidable|from-file]
-jsf-coding-rules [all-rules|shall-rules|shall-will-rules|from-file]
-misra-ac-agc [all-rules|SQ0-subset1|SQ0-subset2|OBL-rules|OBL-REC-rules|
              single-unit-rules|system-decidable-rules|from-file]
-misra-c-2023 [all|all-rules|SQ0-subset1|SQ0-subset2|
              mandatory-required-rules|mandatory|mandatory-required|
              single-unit-rules|system-decidable-rules|from-file]
-misra-c-2023-agc-mode
-misra-cpp [all-rules|SQ0-subset1|SQ0-subset2|required-rules|from-file]
-misra-cpp-2023 [all|all-rules|mandatory-required-rules|mandatory|
                 mandatory-required|single-unit-rules|system-decidable-rules|
```

To check if the Polyspace Access web interface is set up for upload:

1. Navigate again to *polyspaceserverroot*\polyspace\bin.
2. Enter:

```
polyspace-access -host hostName -port portNumber -create-project testProject
```

Here, *hostName* is the name of the server hosting the Polyspace Access web server. For a locally hosted server, use localhost. The *portNumber* is the optional port number of the server. If you omit the port number, 9443 is used.

If the setup was complete, a project called testProject is created in the Polyspace Access web interface.

You are prompted for your login and password each time that you use the polyspace-access command.

3. In a web browser, open this URL: <https://hostName:portNumber/metrics/index.html> Here, *hostName* and *portNumber* are the host name and port number from the previous step.

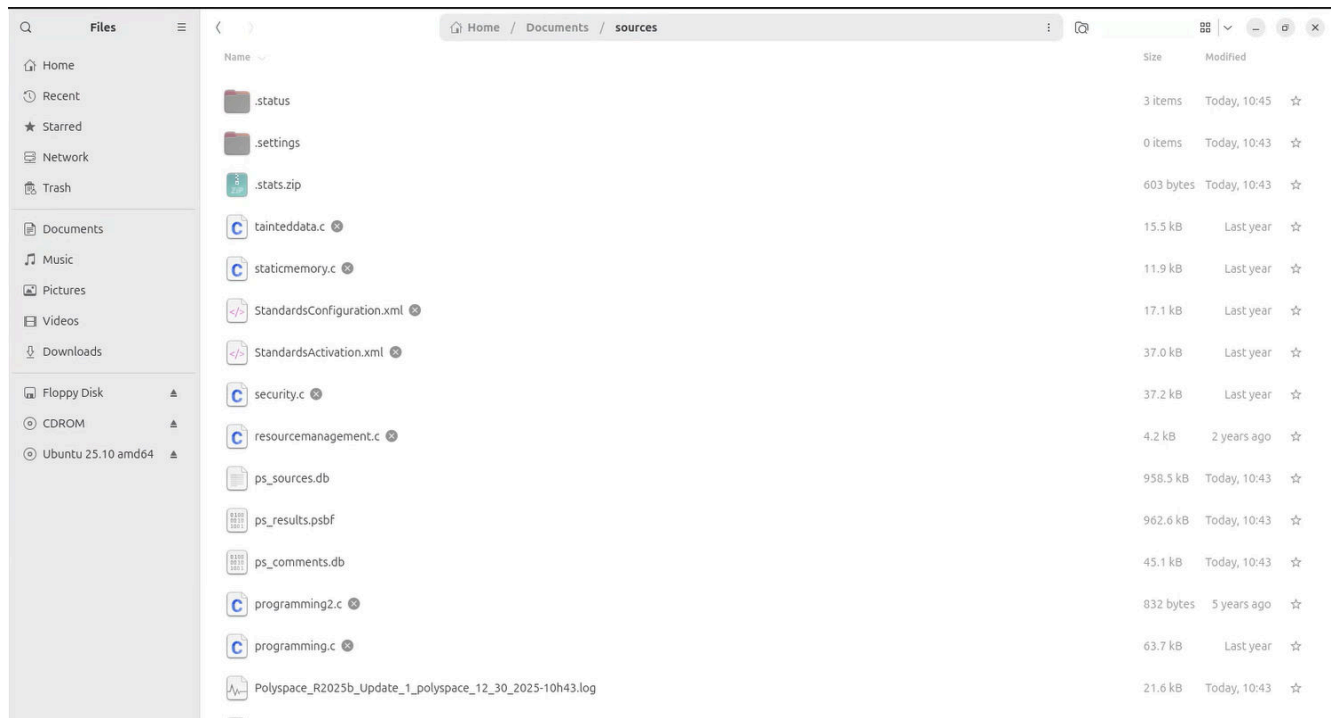
In the **Project Explorer** pane on the Polyspace Access web interface, you see the newly created project testProject.

Run Bug Finder on Sample Files

Try out the commands on sample files provided with your Polyspace installation.

1. Copy the sample source files from
polyspaceserverroot\polyspace\examples\cxx\Bug_Finder_Example\sources to another

folder where you have write permissions. Navigate to this folder at the command line.



2. Enter:

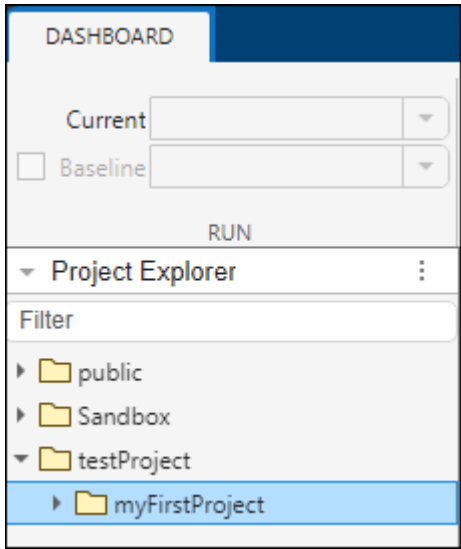
```
polyspace-bug-finder-server -sources numerical.c,dataflow.c -I . -checkers numerical,data_flow
-results-dir .
polyspace-access -host hostName -port portNumber -protocol http -create-project testProject
polyspace-access -host hostName -port portNumber -protocol http -upload . -project myFirstProject
-parent-project testProject
```

```
*****
*** Bug-finder - Module Analysis done
***
*****
Ending at: 2025-12-30 10:43:30
User time for module: 00:00:00.49 (0.49real, 0.35u + 0.14s (0.02gc))
User time for polyspace-bug-finder-server: 00:00:07.34 (7.34real, 12.37u + 9.15s (0.06gc))
***
*** End of Polyspace analysis
***

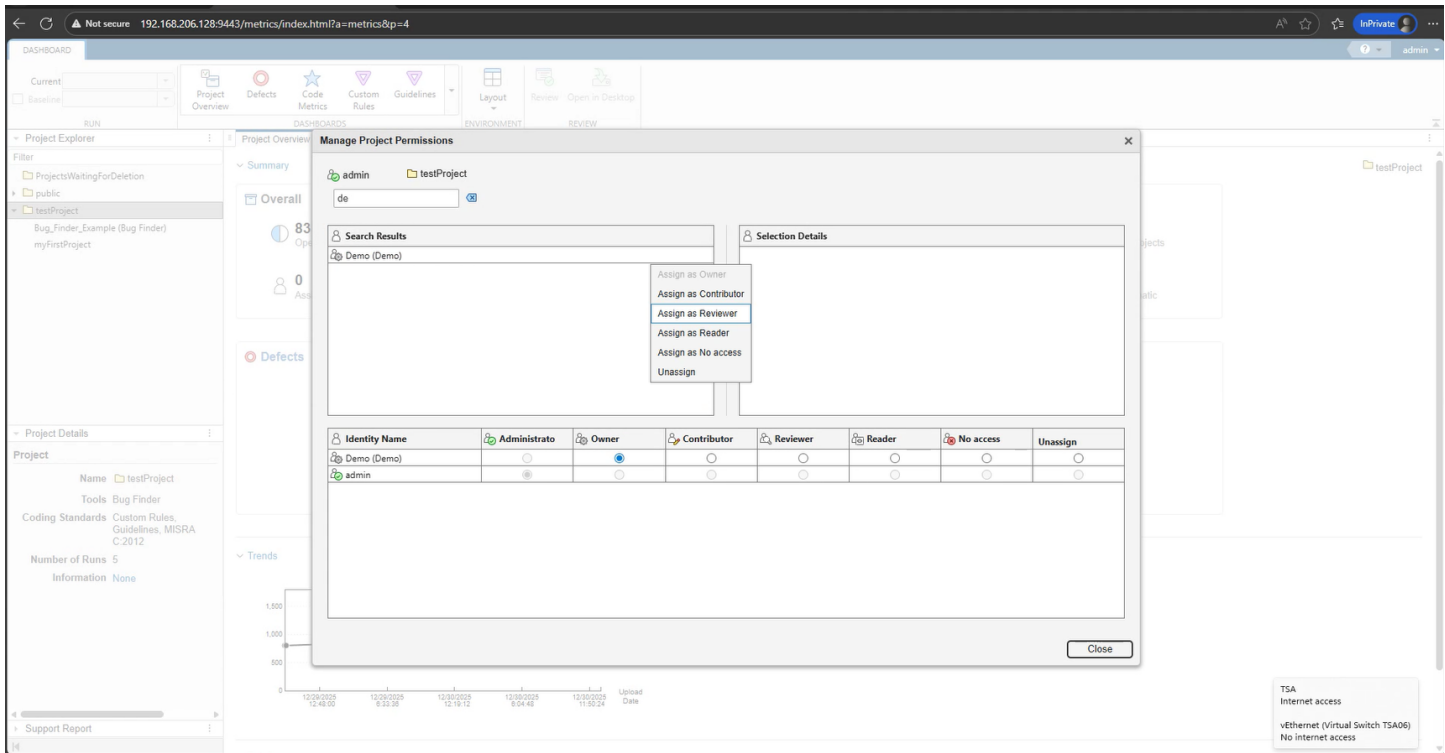
Trial License -- for use to evaluate programs for possible purchase as an end-user only.

polyspace-server-06@polyspace-server-06-VMware-Virtual-Platform:~/Documents/sources$ polyspace-access -host localhost -port 9443 -protocol http -create-project testProject
login: admin
password:
Connecting to http://localhost:9443
Connecting as admin
PROJECT_EXISTS testProject
PROJECT_ID 4
Command Completed
polyspace-server-06@polyspace-server-06-VMware-Virtual-Platform:~/Documents/sources$ polyspace-access -host localhost -port 9443 -protocol http -upload . -project myFirstProject -parent-project testProject
login: admin
password:
Connecting to http://localhost:9443
Connecting as admin
Zipping results
Upload . in testProject/myFirstProject
Upload with IMPORT_ID 1766972151708_0f182fa2-7536-479f-8259-3bc150855474.zip
Uploading Progress: 0% . In Queue. Upload will begin once other upload(s) complete...
Uploading RunId: 6 Progress: 50%
Uploading RunId: 6 Progress: 82%
Uploading RunId: 6 Progress: 98%
Upload successful for RUN_ID 6 and PROJECT_ID 6
You can now view the results via your web browser at
ACCESS_URL http://localhost:9443/metrics/index.html?a=metrics&p=6&r=6
Upload completed for 1 / 1 runs
Command Completed
```

Refresh the Polyspace Access web interface. You see a folder testProject on the **Project Explorer** pane. The folder contains one project myFirstProject.



If it is not available on user end, login to Cluster Dashboard using admin credentials and assign it to the end user.



Dashboard

Import ▾

Create ▾

Filter ▾

Filter by ID or display name...

Showing 1-2 of 2

Previous

Next

ID	Type	Display Name	Source	Actions
admin ADMIN	user		internal	✕ ▾
Demo	user	Demo	internal	✕ ▾

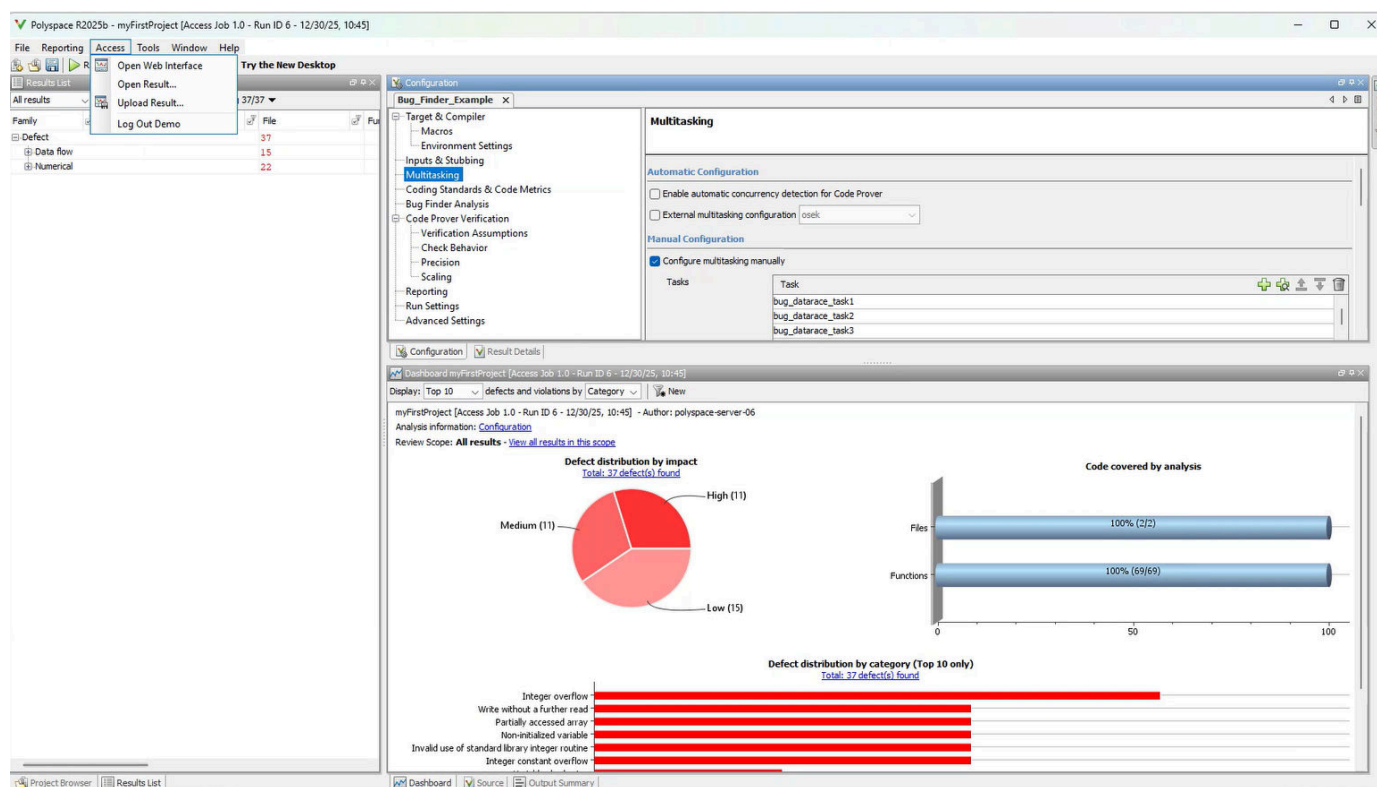
Showing 1-2 of 2

Previous

Next

More info: https://www.mathworks.com/help/polyspace_access/ug/manage-users-and-view-project-trends.html?s_tid=srchtitle_support_results_9_polyspace%2520access%2520permission

On Windows, open Polyspace R2025b, upload result



Appendix

Similar workflow can be used for Polyspace Code Prover: <https://www.mathworks.com/help/codeprover/gs/run-code-prover-on-server.html>